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Aim:

To implement a Hyperledger Fabric network and demonstrate asset management using chaincode.

Theory:

Hyperledger Fabric is a permissioned and modular blockchain platform designed for enterprise applications. Unlike public blockchains, participants in a Fabric network are known and authorized. The network consists of organizations, peers, ordering services, and channels. A channel provides a private communication and transaction environment between participating organizations.

In Hyperledger Fabric, business logic is implemented using chaincode, which is similar to a smart contract. Chaincode defines how assets are created, accessed, and modified on the blockchain ledger. In this experiment, the basic chaincode is used for managing digital assets.

The ledger in Fabric maintains the current state of assets in the world state along with the transaction history. Each asset contains attributes such as ID, color, size, owner, and appraised value. Chaincode functions such as CreateAsset, ReadAsset, UpdateAsset, TransferAsset, DeleteAsset, and GetAllAssets are used to perform different asset-management operations.

A transaction is proposed by a client and endorsed by the required peers. After successful endorsement, it is sent to the ordering service, which orders transactions into blocks and distributes them to the peers. The peers then validate and commit the transactions to their ledgers. This process provides controlled access, consistency, and transparency while maintaining the permissioned nature of the network.

Procedure:

1. Set up the Hyperledger Fabric test network.
2. Create the channel named mychannel.
3. Deploy the Go-based basic chaincode on the channel.
4. Initialize the ledger using the InitLedger function.
5. Read and display existing assets using ReadAsset and GetAllAssets.
6. Create a new asset named asset10 using the CreateAsset function.
7. Transfer ownership of asset10 to Alice using the TransferAsset function.
8. Update the properties of asset10 using the UpdateAsset function.

9. Delete asset10 using the DeleteAsset function and verify its deletion using AssetExists.

Output:

Fabric network and channel successfully created.

```
1.5.17: Pulling from hyperledger/fabric-ca
18afb8c8d2a4: Pull complete
6f4ebca3e823: Pull complete
b9649d3f054a: Pull complete
0b966e606d9a: Pull complete
c36bacd85367: Download complete
Digest: sha256:453a0a727e82c4960b797e379adc231e853ee23ae7526db53afef77b5074117e
Status: Downloaded newer image for ghcr.io/hyperledger/fabric-ca:1.5.17
ghcr.io/hyperledger/fabric-ca:1.5.17
===> List out hyperledger images
ghcr.io/hyperledger/fabric-baseos:2.5.16    109b4a15b282        257MB         83.2MB
ghcr.io/hyperledger/fabric-ca:1.5.17        453a0a727e82        381MB          123MB
ghcr.io/hyperledger/fabric-ccenv:2.5.16     807493f095f6       1.01GB          257MB
ghcr.io/hyperledger/fabric-orderer:2.5.16   1d918fc7e1e0        182MB         50.7MB
ghcr.io/hyperledger/fabric-peer:2.5.16      58979b65744b        232MB         66.8MB
hyperledger/fabric-baseos:2.5               109b4a15b282        257MB         83.2MB
hyperledger/fabric-baseos:2.5.16            109b4a15b282        257MB         83.2MB
hyperledger/fabric-baseos:latest            109b4a15b282        257MB         83.2MB
hyperledger/fabric-ca:1.5                   453a0a727e82        381MB          123MB
hyperledger/fabric-ca:1.5.17                453a0a727e82        381MB          123MB
hyperledger/fabric-ca:latest                453a0a727e82        381MB          123MB
hyperledger/fabric-ccenv:2.5                807493f095f6       1.01GB          257MB
hyperledger/fabric-ccenv:2.5.16             807493f095f6       1.01GB          257MB
hyperledger/fabric-ccenv:latest             807493f095f6       1.01GB          257MB
hyperledger/fabric-orderer:2.5              1d918fc7e1e0        182MB         50.7MB
hyperledger/fabric-orderer:2.5.16           1d918fc7e1e0        182MB         50.7MB
hyperledger/fabric-orderer:latest           1d918fc7e1e0        182MB         50.7MB
hyperledger/fabric-peer:2.5                 58979b65744b        232MB         66.8MB
hyperledger/fabric-peer:2.5.16              58979b65744b        232MB         66.8MB
hyperledger/fabric-peer:latest              58979b65744b        232MB         66.8MB
```

Basic chaincode successfully deployed.

```
Using docker and docker-compose
Generating channel genesis block 'mychannel.block'
Using organization 1
/home/chirag/hyperledger/fabric-samples/bin/configtxgen
+ '[' 0 -eq 1 ']'
+ configtxgen -profile ChannelUsingRaft -outputBlock ./channel-artifacts/mychannel.block -channelID mychannel
2026-10-07 10:12:59.658 IST 0001 INFO [common.tools.configtxgen] main -> Loading configuration
2026-10-07 10:12:59.663 IST 0002 INFO [common.tools.configtxgen.localconfig] completeInitialization -> orderer type: etcdraft
2026-10-07 10:12:59.663 IST 0003 INFO [common.tools.configtxgen.localconfig] completeInitialization -> Orderer.EtcdRaft.Options u
nset, setting to tick_interval:"500ms" election_tick:10 heartbeat_tick:1 max_inflight_blocks:5 snapshot_interval_size:16777216
2026-10-07 10:12:59.663 IST 0004 INFO [common.tools.configtxgen.localconfig] Load -> Loaded configuration: /home/chirag/hyperledg
er/fabric-samples/test-network/configtx/configtx.yaml
2026-10-07 10:12:59.664 IST 0005 INFO [common.tools.configtxgen] doOutputBlock -> Generating genesis block
2026-10-07 10:12:59.664 IST 0006 INFO [common.tools.configtxgen] doOutputBlock -> Creating application channel genesis block
2026-10-07 10:12:59.666 IST 0007 INFO [common.tools.configtxgen] doOutputBlock -> Writing genesis block
+ res=0
Creating channel mychannel
Adding orderers
+ . scripts/orderer.sh mychannel
+ '[' 0 -eq 1 ']'
+ res=0
Status: 201
{
        "name": "mychannel",
        "url": "/participation/v1/channels/mychannel",
        "consensusRelation": "consenter",
        "status": "active",
        "height": 1
}

Channel 'mychannel' created


ig.json --type common.Config --output /home/chirag/hyperledger/fabric-samples/test-network/channel-artifacts/modified_config.pb
++ configtxlator compute_update --channel_id mychannel --original /home/chirag/hyperledger/fabric-samples/test-network/channel-ar
tifacts/original_config.pb --updated /home/chirag/hyperledger/fabric-samples/test-network/channel-artifacts/modified_config.pb --
output /home/chirag/hyperledger/fabric-samples/test-network/channel-artifacts/config_update.pb
++ configtxlator proto_decode --input /home/chirag/hyperledger/fabric-samples/test-network/channel-artifacts/config_update.pb --t
ype common.ConfigUpdate --output /home/chirag/hyperledger/fabric-samples/test-network/channel-artifacts/config_update.json
++ jq .
+++ cat /home/chirag/hyperledger/fabric-samples/test-network/channel-artifacts/config_update.json
++ echo '{"payload":{"header":{"channel_header":{"channel_id":"mychannel", "type":2}},"data":{"config_update":{' '"channel_id":'
'"mychannel",' '"isolated_data":' '{},' '"read_set":' '{' '"groups":' '{' '"Application":' '{' '"groups":' '{' '"Org2MSP":' '{' '
"groups":' '{},' '"mod_policy":' '"",' '"policies":' '{' '"Admins":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0
"' '},' '"Endorsement":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '},' '"Readers":' '{' '"mod_policy":' '"
,' '"policy":' null, '"version":' '"0"' '},' '"Writers":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '}' '},'
 '"values":' '{' '"MSP":' '{' '"mod_policy":' '"",' '"value":' null, '"version":' '"0"' '}' '},' '"version":' '"0"' '}' '},' '"mo
d_policy":' '"",' '"policies":' '{},' '"values":' '{},' '"version":' '"0"' '}' '},' '"mod_policy":' '"",' '"policies":' '{},' '"v
alues":' '{},' '"version":' '"0"' '},' '"write_set":' '{' '"groups":' '{' '"Application":' '{' '"groups":' '{' '"Org2MSP":' '{' '
"groups":' '{},' '"mod_policy":' '"Admins",' '"policies":' '{' '"Admins":' '{' '"mod_policy":' '"",' '"policy":' null, '"version"
:' '"0"' '},' '"Endorsement":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '},' '"Readers":' '{' '"mod_policy"
:' '"",' '"policy":' null, '"version":' '"0"' '},' '"Writers":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '}
' '},' '"values":' '{' '"AnchorPeers":' '{' '"mod_policy":' '"Admins",' '"value":' '{' '"anchor_peers":' '[' '{' '"host":' '"peer
0.org2.example.com",' '"port":' 9051 '}' ']' '},' '"version":' '"0"' '},' '"MSP":' '{' '"mod_policy":' '"",' '"value":' null, '"v
ersion":' '"0"' '}' '},' '"version":' '"1"' '}' '},' '"mod_policy":' '"",' '"policies":' '{},' '"values":' '{},' '"version":' '"0
"' '}' '},' '"mod_policy":' '"",' '"policies":' '{},' '"values":' '{},' '"version":' '"0"' '}' '}}}}'
++ configtxlator proto_encode --input /home/chirag/hyperledger/fabric-samples/test-network/channel-artifacts/config_update_in_env
elope.json --type common.Envelope --output /home/chirag/hyperledger/fabric-samples/test-network/channel-artifacts/Org2MSPanchors.
tx
2026-10-07 10:13:09.448 IST 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-10-07 10:13:09.464 IST 0002 INFO [channelCmd] update -> Successfully submitted channel update
Anchor peer set for org 'Org2MSP' on channel 'mychannel'
Channel 'mychannel' joined
➜ hyperledger/fabric-samples/test-network  main ❯
```

Docker containers running.

```
➜ hyperledger/fabric-samples/test-network  main ❯
docker ps                                                                                                                  10:13
CONTAINER ID   IMAGE                                COMMAND             CREATED              STATUS              PORTS
                                                                                                                  NAMES
6ff9dace2c4a   hyperledger/fabric-peer:latest       "peer node start"   About a minute ago   Up About a minute   0.0.0.0:7051->705
1/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp                                                    peer0.org1
.example.com
0eb738ae2486   hyperledger/fabric-peer:latest       "peer node start"   About a minute ago   Up About a minute   0.0.0.0:9051->905
1/tcp, [::]:9051->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp                                          peer0.org2
.example.com
c57c8f3f3827   hyperledger/fabric-orderer:latest    "orderer"           About a minute ago   Up About a minute   0.0.0.0:7050->705
0/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp   orderer.ex
ample.com
```

InitLedger transaction successful.

```
Checking the commit readiness of the chaincode definition successful on peer0.org2 on channel 'mychannel'
Using organization 1
Using organization 2
+ peer lifecycle chaincode commit -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile /home/chirag/
hyperledger/fabric-samples/test-network/organizations/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem --channel
ID mychannel --name basic --peerAddresses localhost:7051 --tlsRootCertFiles /home/chirag/hyperledger/fabric-samples/test-network/
organizations/peerOrganizations/org1.example.com/tlsca/tlsca.org1.example.com-cert.pem --peerAddresses localhost:9051 --tlsRootCe
rtFiles /home/chirag/hyperledger/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.ex
ample.com-cert.pem --version 1.0 --sequence 1
+ res=0
2026-10-07 10:17:19.864 IST 0001 INFO [chaincodeCmd] ClientWait -> txid [0c142e577cb8b4eba0a417e3d2f9ed7700cb6dfffec63ebd354ae5ea
2ac877ca] committed with status (VALID) at localhost:7051
2026-10-07 10:17:19.871 IST 0002 INFO [chaincodeCmd] ClientWait -> txid [0c142e577cb8b4eba0a417e3d2f9ed7700cb6dfffec63ebd354ae5ea
2ac877ca] committed with status (VALID) at localhost:9051
Chaincode definition committed on channel 'mychannel'
Using organization 1
Querying chaincode definition on peer0.org1 on channel 'mychannel'...
Attempting to Query committed status on peer0.org1, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org1 on channel 'mychannel'
Using organization 2
Querying chaincode definition on peer0.org2 on channel 'mychannel'...
Attempting to Query committed status on peer0.org2, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org2 on channel 'mychannel'
```

```
hyperledger/fabric-samples/test-network  main
peer chaincode invoke \                                                                                    10:26
      -o localhost:7050 \
      --ordererTLSHostnameOverride orderer.example.com \
      --tls \
      --cafile "$PWD/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com
-cert.pem" \
      -C mychannel \
      -n basic \
      --peerAddresses localhost:7051 \
      --tlsRootCertFiles "$PWD/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt" \
      --peerAddresses localhost:9051 \
      --tlsRootCertFiles "$PWD/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt" \
      -c '{"function":"InitLedger","Args":[]}'
2026-10-07 10:27:17.659 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
```

Existing asset retrieved using ReadAsset / GetAllAssets.

```
hyperledger/fabric-samples/test-network  main
peer chaincode query \
      -C mychannel \
      -n basic \
      -c '{"function":"ReadAsset","Args":["asset1"]}'
{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5}
```

```
hyperledger/fabric-samples/test-network  main
peer chaincode query \                                                                                     10:27
      -C mychannel \
      -n basic \
      -c '{"function":"GetAllAssets","Args":[]}'
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5},{"AppraisedValue":400,"Color":"red","ID":"asset2",
"Owner":"Brad","Size":5},{"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10},{"AppraisedValue":600,"
Color":"yellow","ID":"asset4","Owner":"Max","Size":10},{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Siz
e":15},{"AppraisedValue":800,"Color":"white","ID":"asset6","Owner":"Michel","Size":15}]
```

Asset10 successfully created and displayed.

```
hyperledger/fabric-samples/test-network  main
peer chaincode invoke \                                                                                    10:29
      -o localhost:7050 \
      --ordererTLSHostnameOverride orderer.example.com \
      --tls \
      --cafile "$PWD/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com
-cert.pem" \
      -C mychannel \
      -n basic \
      --peerAddresses localhost:7051 \
      --tlsRootCertFiles "$PWD/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt" \
      --peerAddresses localhost:9051 \
      --tlsRootCertFiles "$PWD/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt" \
      -c '{"function":"CreateAsset","Args":["asset10","red","10","Chirag","1000"]}'
2026-10-07 10:30:05.428 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
```

```
hyperledger/fabric-samples/test-network  main
peer chaincode query \
      -C mychannel \
      -n basic \
      -c '{"function":"ReadAsset","Args":["asset10"]}'
{"AppraisedValue":1000,"Color":"red","ID":"asset10","Owner":"Chirag","Size":10}
```

Ownership of asset10 transferred to alice.

```
hyperledger/fabric-samples/test-network  main
peer chaincode invoke \                                                                                    10:34
      -o localhost:7050 \
      --ordererTLSHostnameOverride orderer.example.com \
      --tls \
      --cafile "$PWD/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com
-cert.pem" \
      -C mychannel \
      -n basic \
      --peerAddresses localhost:7051 \
      --tlsRootCertFiles "$PWD/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt" \
      --peerAddresses localhost:9051 \
      --tlsRootCertFiles "$PWD/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt" \
      -c '{"function":"TransferAsset","Args":["asset10","Alice"]}'
2026-10-07 10:34:02.384 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 pa
yload:"Chirag"
hyperledger/fabric-samples/test-network  main
peer chaincode query \                                                                                     10:34
      -C mychannel \
      -n basic \
      -c '{"function":"ReadAsset","Args":["asset10"]}'
{"AppraisedValue":1000,"Color":"red","ID":"asset10","Owner":"Alice","Size":10}
```

Asser10 successfully updated.

```
hyperledger/fabric-samples/test-network  main
peer chaincode invoke \                                                                                    10:34
      -o localhost:7050 \
      --ordererTLSHostnameOverride orderer.example.com \
      --tls \
      --cafile "$PWD/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com
-cert.pem" \
      -C mychannel \
      -n basic \
      --peerAddresses localhost:7051 \
      --tlsRootCertFiles "$PWD/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt" \
      --peerAddresses localhost:9051 \
      --tlsRootCertFiles "$PWD/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt" \
      -c '{"function":"UpdateAsset","Args":["asset10","blue","15","Alice","1500"]}'
2026-10-07 10:34:39.413 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
hyperledger/fabric-samples/test-network  main
peer chaincode query \                                                                                     10:34
      -C mychannel \
      -n basic \
      -c '{"function":"ReadAsset","Args":["asset10"]}'
{"AppraisedValue":1500,"Color":"blue","ID":"asset10","Owner":"Alice","Size":15}
```

AssetExist returned false after deletion.

```
hyperledger/fabric-samples/test-network  main
peer chaincode invoke \                                                                                    10:34
      -o localhost:7050 \
      --ordererTLSHostnameOverride orderer.example.com \
      --tls \
      --cafile "$PWD/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.co
-cert.pem" \
      -C mychannel \
      -n basic \
      --peerAddresses localhost:7051 \
      --tlsRootCertFiles "$PWD/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt" \
      --peerAddresses localhost:9051 \
      --tlsRootCertFiles "$PWD/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt" \
      -c '{"function":"DeleteAsset","Args":["asset10"]}'
2026-10-07 10:35:07.021 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
hyperledger/fabric-samples/test-network  main
peer chaincode query \                                                                                     10:35
      -C mychannel \
      -n basic \
      -c '{"function":"AssetExists","Args":["asset10"]}'
false
```

Conclusion:

The Hyperledger Fabric network was successfully implemented, and the chaincode was used to perform asset creation, retrieval, ownership transfer, updating, and deletion operations. The successful deletion of asset10 was verified using the AssetExists function, which returned false. Thus, the experiment demonstrated asset management using chaincode on a permissioned blockchain network.